

# APM1220: Applied Multivariate Data Analysis

## Formative Assessment 1

### Problem 2.2 (5 points)

Given

$$A = \begin{bmatrix} -1 & 3 \\ 4 & 2 \end{bmatrix}, \quad B = \begin{bmatrix} 4 & -3 \\ 1 & -2 \\ -2 & 0 \end{bmatrix}, \quad C = \begin{bmatrix} 5 \\ -4 \\ 2 \end{bmatrix}.$$

(a)  $5A$

$$5A = \begin{bmatrix} -5 & 15 \\ 20 & 10 \end{bmatrix}$$

(b)  $BA$

$B$  is  $3 \times 2$  and  $A$  is  $2 \times 2$ , so  $BA$  is  $3 \times 2$  (dimensions conform:  $2 = 2$ ).

$$BA = \begin{bmatrix} 4 & -3 \\ 1 & -2 \\ -2 & 0 \end{bmatrix} \begin{bmatrix} -1 & 3 \\ 4 & 2 \end{bmatrix} = \begin{bmatrix} 4(-1) + (-3)(4) & 4(3) + (-3)(2) \\ 1(-1) + (-2)(4) & 1(3) + (-2)(2) \\ -2(-1) + 0(4) & -2(3) + 0(2) \end{bmatrix} = \begin{bmatrix} -16 & 6 \\ -9 & -1 \\ 2 & -6 \end{bmatrix}$$

(c)  $A'B'$

$$A' = \begin{bmatrix} -1 & 4 \\ 3 & 2 \end{bmatrix}, \quad B' = \begin{bmatrix} 4 & 1 & -2 \\ -3 & -2 & 0 \end{bmatrix}$$

$A'$  is  $2 \times 2$ ,  $B'$  is  $2 \times 3$ , so  $A'B'$  is  $2 \times 3$  (dimensions conform:  $2 = 2$ ). Note  $A'B' = (BA)'$ .

$$A'B' = \begin{bmatrix} -1 & 4 \\ 3 & 2 \end{bmatrix} \begin{bmatrix} 4 & 1 & -2 \\ -3 & -2 & 0 \end{bmatrix} = \begin{bmatrix} -1(4) + 4(-3) & -1(1) + 4(-2) & -1(-2) + 4(0) \\ 3(4) + 2(-3) & 3(1) + 2(-2) & 3(-2) + 2(0) \end{bmatrix} = \begin{bmatrix} -16 & -9 & 2 \\ 6 & -1 & -6 \end{bmatrix}$$

which indeed equals  $(BA)'$  computed in (b), as required.

(d)  $C'B$

$C'$  is  $1 \times 3$  and  $B$  is  $3 \times 2$ , so  $C'B$  is  $1 \times 2$  (dimensions conform:  $3 = 3$ ).

$$C'B = [5 \quad -4 \quad 2] \begin{bmatrix} 4 & -3 \\ 1 & -2 \\ -2 & 0 \end{bmatrix} = [5(4) + (-4)(1) + 2(-2) \quad 5(-3) + (-4)(-2) + 2(0)] = [12 \quad -7]$$

(e) Is  $AB$  defined?

$A$  is  $2 \times 2$  and  $B$  is  $3 \times 2$ . The number of columns of  $A$  (2) does not equal the number of rows of  $B$  (3), so the product  $AB$  is **not defined**.

### Problem 2.4 (6 points)

(a)  $(A')^{-1} = (A^{-1})'$

Since  $A^{-1}$  exists,  $AA^{-1} = I$ . Taking the transpose of both sides and using  $I' = I$ ,

$$(AA^{-1})' = I' = I.$$

By the transpose-of-a-product rule,  $(AA^{-1})' = (A^{-1})'A'$ , so

$$(A^{-1})'A' = I.$$

This equation states that  $(A^{-1})'$  is a matrix which, when multiplied on the right by  $A'$ , gives the identity; hence  $(A^{-1})'$  is the (unique) inverse of  $A'$ . Therefore

$$(A')^{-1} = (A^{-1})'. \quad \blacksquare$$

(b)  $(AB)^{-1} = B^{-1}A^{-1}$

Using associativity of matrix multiplication and  $A^{-1}A = I$ ,  $B^{-1}B = I$ :

$$(B^{-1}A^{-1})(AB) = B^{-1}(A^{-1}A)B = B^{-1}(I)B = B^{-1}B = I.$$

Likewise,

$$(AB)(B^{-1}A^{-1}) = A(BB^{-1})A^{-1} = A(I)A^{-1} = AA^{-1} = I.$$

Since  $B^{-1}A^{-1}$  multiplied by  $AB$  on either side yields the identity matrix,  $B^{-1}A^{-1}$  is the (unique) inverse of  $AB$ . Therefore

$$(AB)^{-1} = B^{-1}A^{-1}. \quad \blacksquare$$

### Problem 2.8 (6 points)

Given

$$A = \begin{bmatrix} 1 & 2 \\ 2 & -2 \end{bmatrix}.$$

#### Eigenvalues

Solve  $|A - \lambda I| = 0$ :

$$\det \begin{bmatrix} 1 - \lambda & 2 \\ 2 & -2 - \lambda \end{bmatrix} = (1 - \lambda)(-2 - \lambda) - 4 = \lambda^2 + \lambda - 6 = 0$$

$$(\lambda - 2)(\lambda + 3) = 0 \implies \lambda_1 = 2, \quad \lambda_2 = -3.$$

(Check:  $\lambda_1 + \lambda_2 = -1 = \text{tr}(A)$ , and  $\lambda_1\lambda_2 = -6 = |A|$ .)

## Eigenvectors

**For  $\lambda_1 = 2$ :** Solve  $(A - 2I)v = 0$ :

$$\begin{bmatrix} -1 & 2 \\ 2 & -4 \end{bmatrix} \begin{bmatrix} v_1 \\ v_2 \end{bmatrix} = 0 \implies v_1 = 2v_2.$$

Take  $v = (2, 1)'$ ; its length is  $\sqrt{2^2 + 1^2} = \sqrt{5}$ , so the normalized eigenvector is

$$e_1 = \frac{1}{\sqrt{5}} \begin{bmatrix} 2 \\ 1 \end{bmatrix}.$$

**For  $\lambda_2 = -3$ :** Solve  $(A + 3I)v = 0$ :

$$\begin{bmatrix} 4 & 2 \\ 2 & 1 \end{bmatrix} \begin{bmatrix} v_1 \\ v_2 \end{bmatrix} = 0 \implies v_2 = -2v_1.$$

Take  $v = (1, -2)'$ ; its length is  $\sqrt{5}$ , so

$$e_2 = \frac{1}{\sqrt{5}} \begin{bmatrix} 1 \\ -2 \end{bmatrix}.$$

(As expected for a symmetric matrix with distinct eigenvalues,  $e_1'e_2 = \frac{1}{5}(2 - 2) = 0$ .)

## Spectral decomposition

By (2-16),  $A = \lambda_1 e_1 e_1' + \lambda_2 e_2 e_2'$ :

$$e_1 e_1' = \frac{1}{5} \begin{bmatrix} 4 & 2 \\ 2 & 1 \end{bmatrix}, \quad e_2 e_2' = \frac{1}{5} \begin{bmatrix} 1 & -2 \\ -2 & 4 \end{bmatrix}$$

$$A = 2 \left( \frac{1}{5} \begin{bmatrix} 4 & 2 \\ 2 & 1 \end{bmatrix} \right) + (-3) \left( \frac{1}{5} \begin{bmatrix} 1 & -2 \\ -2 & 4 \end{bmatrix} \right) = \frac{1}{5} \begin{bmatrix} 8 & 4 \\ 4 & 2 \end{bmatrix} + \frac{1}{5} \begin{bmatrix} -3 & 6 \\ 6 & -12 \end{bmatrix} = \begin{bmatrix} 1 & 2 \\ 2 & -2 \end{bmatrix}$$

which recovers  $A$ , confirming the decomposition.

## Problem 2.11 (5 points)

**Claim:** For a  $p \times p$  diagonal matrix  $A = \{a_{ij}\}$  with  $a_{ij} = 0$  for  $i \neq j$ ,

$$|A| = a_{11}a_{22} \cdots a_{pp}.$$

**Proof (by induction on  $p$ ).**

*Base case* ( $p = 1$ ):  $A = [a_{11}]$ , and  $|A| = a_{11}$  trivially.

*Inductive step:* Assume the result holds for  $(p-1) \times (p-1)$  diagonal matrices. By Definition 2A.24, the determinant of  $A$  expanded by cofactors along the first row is

$$|A| = a_{11}A_{11} + a_{12}A_{12} + \cdots + a_{1p}A_{1p},$$

where  $A_{1j}$  is the cofactor of  $a_{1j}$ . Since  $A$  is diagonal,  $a_{1j} = 0$  for all  $j \neq 1$ , so every term except the first vanishes:

$$|A| = a_{11}A_{11}.$$

The cofactor  $A_{11} = (-1)^{1+1}|M_{11}| = |M_{11}|$ , where  $M_{11}$  is the submatrix obtained by deleting the first row and first column of  $A$ . Because  $A$  is diagonal,  $M_{11}$  is exactly the  $(p-1) \times (p-1)$  diagonal matrix with diagonal entries  $a_{22}, \dots, a_{pp}$ . By the induction hypothesis,

$$|M_{11}| = a_{22}a_{33} \cdots a_{pp}.$$

Therefore

$$|A| = a_{11}(a_{22}a_{33} \cdots a_{pp}) = a_{11}a_{22} \cdots a_{pp}.$$

By induction, the result holds for all  $p \geq 1$ . ■

## Problem 2.24 (5 points)

Given

$$\Sigma = \begin{bmatrix} 4 & 0 & 0 \\ 0 & 9 & 0 \\ 0 & 0 & 1 \end{bmatrix}.$$

(a)  $\Sigma^{-1}$

Since  $\Sigma$  is diagonal, its inverse is the diagonal matrix of reciprocals:

$$\Sigma^{-1} = \begin{bmatrix} \frac{1}{4} & 0 & 0 \\ 0 & \frac{1}{9} & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

(b) Eigenvalues and eigenvectors of  $\Sigma$

For a diagonal matrix, the eigenvalues are the diagonal entries themselves, with the standard basis vectors as the corresponding (already normalized) eigenvectors. Ordering the eigenvalues from largest to smallest:

$$\lambda_1 = 9, e_1 = \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}; \quad \lambda_2 = 4, e_2 = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}; \quad \lambda_3 = 1, e_3 = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}.$$

(c) Eigenvalues and eigenvectors of  $\Sigma^{-1}$

If  $\Sigma e = \lambda e$ , then  $\Sigma^{-1}e = \lambda^{-1}e$ , so  $\Sigma^{-1}$  has the *same* eigenvectors as  $\Sigma$ , with reciprocal eigenvalues. Ordered largest to smallest:

$$\lambda_1 = 1, e_1 = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}; \quad \lambda_2 = \frac{1}{4}, e_2 = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}; \quad \lambda_3 = \frac{1}{9}, e_3 = \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}.$$

**Problem 2.30 (8 points)**

Given  $X' = [X_1, X_2, X_3, X_4]$ ,  $\mu'_X = [4, 3, 2, 1]$ , and

$$\Sigma_X = \begin{bmatrix} 3 & 0 & 2 & 2 \\ 0 & 1 & 1 & 0 \\ 2 & 1 & 9 & -2 \\ 2 & 0 & -2 & 4 \end{bmatrix},$$

partitioned as  $X = \begin{bmatrix} X^{(1)} \\ X^{(2)} \end{bmatrix}$  with  $X^{(1)} = (X_1, X_2)'$ ,  $X^{(2)} = (X_3, X_4)'$ . Correspondingly,

$$\mu^{(1)} = \begin{bmatrix} 4 \\ 3 \end{bmatrix}, \quad \mu^{(2)} = \begin{bmatrix} 2 \\ 1 \end{bmatrix}, \quad \Sigma_{11} = \begin{bmatrix} 3 & 0 \\ 0 & 1 \end{bmatrix}, \quad \Sigma_{22} = \begin{bmatrix} 9 & -2 \\ -2 & 4 \end{bmatrix}, \quad \Sigma_{12} = \begin{bmatrix} 2 & 2 \\ 1 & 0 \end{bmatrix}.$$

Also  $A = \begin{bmatrix} 1 & 2 \end{bmatrix}$  and  $B = \begin{bmatrix} 1 & -2 \\ 2 & -1 \end{bmatrix}$ .

(a)  $E(X^{(1)})$

$$E(X^{(1)}) = \mu^{(1)} = \begin{bmatrix} 4 \\ 3 \end{bmatrix}$$

(b)  $E(AX^{(1)})$

$$E(AX^{(1)}) = A\mu^{(1)} = \begin{bmatrix} 1 & 2 \end{bmatrix} \begin{bmatrix} 4 \\ 3 \end{bmatrix} = 4 + 6 = 10$$

(c)  $\text{Cov}(X^{(1)})$

$$\text{Cov}(X^{(1)}) = \Sigma_{11} = \begin{bmatrix} 3 & 0 \\ 0 & 1 \end{bmatrix}$$

(d)  $\text{Cov}(AX^{(1)})$

$$\text{Cov}(AX^{(1)}) = A\Sigma_{11}A' = \begin{bmatrix} 1 & 2 \end{bmatrix} \begin{bmatrix} 3 & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 1 \\ 2 \end{bmatrix} = \begin{bmatrix} 3 & 2 \end{bmatrix} \begin{bmatrix} 1 \\ 2 \end{bmatrix} = 3 + 4 = 7$$

(e)  $E(X^{(2)})$

$$E(X^{(2)}) = \mu^{(2)} = \begin{bmatrix} 2 \\ 1 \end{bmatrix}$$

(f)  $E(BX^{(2)})$

$$E(BX^{(2)}) = B\mu^{(2)} = \begin{bmatrix} 1 & -2 \\ 2 & -1 \end{bmatrix} \begin{bmatrix} 2 \\ 1 \end{bmatrix} = \begin{bmatrix} 2 - 2 \\ 4 - 1 \end{bmatrix} = \begin{bmatrix} 0 \\ 3 \end{bmatrix}$$

(g)  $\text{Cov}(X^{(2)})$ 

$$\text{Cov}(X^{(2)}) = \Sigma_{22} = \begin{bmatrix} 9 & -2 \\ -2 & 4 \end{bmatrix}$$

(h)  $\text{Cov}(BX^{(2)})$ 

$$\text{Cov}(BX^{(2)}) = B\Sigma_{22}B'$$

First,

$$B\Sigma_{22} = \begin{bmatrix} 1 & -2 \\ 2 & -1 \end{bmatrix} \begin{bmatrix} 9 & -2 \\ -2 & 4 \end{bmatrix} = \begin{bmatrix} 9+4 & -2-8 \\ 18+2 & -4-4 \end{bmatrix} = \begin{bmatrix} 13 & -10 \\ 20 & -8 \end{bmatrix}$$

Then, with  $B' = \begin{bmatrix} 1 & 2 \\ -2 & -1 \end{bmatrix}$ ,

$$(B\Sigma_{22})B' = \begin{bmatrix} 13 & -10 \\ 20 & -8 \end{bmatrix} \begin{bmatrix} 1 & 2 \\ -2 & -1 \end{bmatrix} = \begin{bmatrix} 13+20 & 26+10 \\ 20+16 & 40+8 \end{bmatrix} = \begin{bmatrix} 33 & 36 \\ 36 & 48 \end{bmatrix}$$

So

$$\text{Cov}(BX^{(2)}) = \begin{bmatrix} 33 & 36 \\ 36 & 48 \end{bmatrix}$$

(i)  $\text{Cov}(X^{(1)}, X^{(2)})$ 

$$\text{Cov}(X^{(1)}, X^{(2)}) = \Sigma_{12} = \begin{bmatrix} 2 & 2 \\ 1 & 0 \end{bmatrix}$$

(j)  $\text{Cov}(AX^{(1)}, BX^{(2)})$ 

$$\text{Cov}(AX^{(1)}, BX^{(2)}) = A\Sigma_{12}B'$$

First,

$$A\Sigma_{12} = \begin{bmatrix} 1 & 2 \end{bmatrix} \begin{bmatrix} 2 & 2 \\ 1 & 0 \end{bmatrix} = \begin{bmatrix} 2+2 & 2+0 \end{bmatrix} = \begin{bmatrix} 4 & 2 \end{bmatrix}$$

Then,

$$(A\Sigma_{12})B' = \begin{bmatrix} 4 & 2 \end{bmatrix} \begin{bmatrix} 1 & 2 \\ -2 & -1 \end{bmatrix} = \begin{bmatrix} 4-4 & 8-2 \end{bmatrix} = \begin{bmatrix} 0 & 6 \end{bmatrix}$$

So

$$\text{Cov}(AX^{(1)}, BX^{(2)}) = \begin{bmatrix} 0 & 6 \end{bmatrix}$$

**Problem 2.41 (5 points)**Given  $X' = [X_1, X_2, X_3, X_4]$ ,  $\mu'_X = [3, 2, -2, 0]$ , and

$$\Sigma_X = \begin{bmatrix} 3 & 0 & 0 & 0 \\ 0 & 3 & 0 & 0 \\ 0 & 0 & 3 & 0 \\ 0 & 0 & 0 & 3 \end{bmatrix} = 3I_4,$$

$$A = \begin{bmatrix} 1 & -1 & 0 & 0 \\ 1 & 1 & -2 & 0 \\ 1 & 1 & 1 & -3 \end{bmatrix}.$$

$A$  is  $3 \times 4$  and  $X$  is  $4 \times 1$ , so  $AX$  is  $3 \times 1$  (dimensions conform:  $4 = 4$ ).

(a)  $E(AX)$

$$E(AX) = A\mu_X = \begin{bmatrix} 1 & -1 & 0 & 0 \\ 1 & 1 & -2 & 0 \\ 1 & 1 & 1 & -3 \end{bmatrix} \begin{bmatrix} 3 \\ 2 \\ -2 \\ 0 \end{bmatrix} = \begin{bmatrix} 3 - 2 \\ 3 + 2 + 4 \\ 3 + 2 - 2 - 0 \end{bmatrix} = \begin{bmatrix} 1 \\ 9 \\ 3 \end{bmatrix}$$

(b)  $\text{Cov}(AX)$

Since  $\Sigma_X = 3I_4$ ,

$$\text{Cov}(AX) = A\Sigma_X A' = 3AA'.$$

Computing  $AA'$  using the rows  $r_1 = (1, -1, 0, 0)$ ,  $r_2 = (1, 1, -2, 0)$ ,  $r_3 = (1, 1, 1, -3)$  of  $A$ :

$$r_1 \cdot r_1 = 2, \quad r_1 \cdot r_2 = 1 - 1 + 0 + 0 = 0, \quad r_1 \cdot r_3 = 1 - 1 + 0 + 0 = 0,$$

$$r_2 \cdot r_2 = 1 + 1 + 4 + 0 = 6, \quad r_2 \cdot r_3 = 1 + 1 - 2 + 0 = 0, \quad r_3 \cdot r_3 = 1 + 1 + 1 + 9 = 12.$$

$$AA' = \begin{bmatrix} 2 & 0 & 0 \\ 0 & 6 & 0 \\ 0 & 0 & 12 \end{bmatrix}$$

$$\text{Cov}(AX) = 3AA' = \begin{bmatrix} 6 & 0 & 0 \\ 0 & 18 & 0 \\ 0 & 0 & 36 \end{bmatrix}$$

(c) Which pairs have zero covariance?

Writing  $Y = AX = (Y_1, Y_2, Y_3)'$  where

$$Y_1 = X_1 - X_2, \quad Y_2 = X_1 + X_2 - 2X_3, \quad Y_3 = X_1 + X_2 + X_3 - 3X_4,$$

the covariance matrix found in (b) is diagonal, so every off-diagonal entry is 0. This means **all three pairs** of linear combinations are uncorrelated (zero covariance):

$$\text{Cov}(Y_1, Y_2) = 0, \quad \text{Cov}(Y_1, Y_3) = 0, \quad \text{Cov}(Y_2, Y_3) = 0.$$